

Machine Learning for Business Analytics

GTSC2143

Course Overview

Instructor: Dr. Lily Bei Luo

AY2025-2026 Semester 1



Bachelor in Information Management and Information System



Ph.D. in Information Systems



Assistant Professor in E-business Management and Information Systems

Teaching Experience and Research Interests

+ 01. Teaching Experience

- Undergraduates Level:
 - Introduction to Business Program in Python
 - Business Software Construction
 - Database Management
 - Business Analytics
 - Management Information Systems
- Master Level:
 - eBusiness System Integration
 - Analytical programming with Python

+ 02. Research Interests

- Natural Language Processing
- Human Computer Interaction
- Fintech
- Social Media



Contact Information

Consulting hours: by appointment

- Office: T1-501-R7
- Email: lilybluo@uic.edu.cn

You may also see contact information on iSpace

Timetable in 2025-2026 Semester 1 -- Dr. Lily Bei Luo					
	Monday	Tuesday	Wednesday	Thursday	Friday
8:00-8:50					
9:00-9:50		CONSULTATION HOUR			CONSULTATION HOUR
10:00-10:50		Database Management T8-301			Machine Learning for Business T4-301
11:00-11:50		Database Management T8-301			Machine Learning for Business T4-301
12:00-12:50		CONSULTATION HOUR			CONSULTATION HOUR
13:00-13:50		Machine Learning for Business T4-301			Database Management T8-301
14:00-14:50		MCP CONSULTATION HOUR			CONSULTATION HOUR
15:00-15:50		MCP CONSULTATION HOUR			CONSULTATION HOUR
16:00-16:50		CONSULTATION HOUR			CONSULTATION HOUR

Objectives of the Course

- Explore the fundamental principles and core concepts of Machine Learning in business analytics contexts.
- Analyze various Machine Learning technologies and how they can be applied to business decision-making, market analysis, and operational optimization.
- Apply essential Machine Learning methodologies, tools, and codes to solve real-world business problems, individually and as a group.

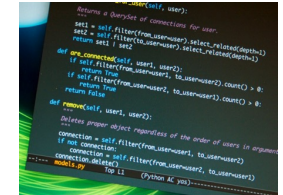
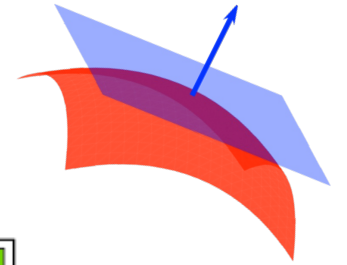
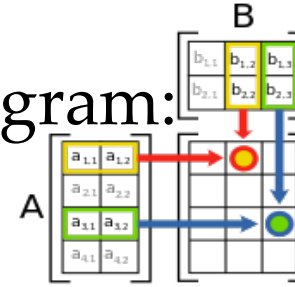
Level of the specialization

Motto: tough concepts made intuitive and applicable

- minimize prerequisite knowledge
- maximize ability to develop and deploy
- learn concepts through case studies

Prerequisites. What do you need to know now?

- You should know how to do math and how to program:
 - Calculus (concept of derivatives)
 - Linear algebra (matrices and vectors)
 - Programming:
 - You will implement some of the algorithms and apply them to datasets
 - Python is the basic programming language.
 - You are expected to create virtual environments, utilize third-party libraries, and build projects via Python.



Teaching Schedule

Week 1	Course Overview Introduction to Machine Learning Python Basics
Week 2	Python and Libraries Data Loading Data Preprocessing
Week 3	Data Visualization
Week 4	Regression I (House Pricing prediction)
Week 5	Regression II (House Pricing prediction)
Week 6	Classification I (Sentiment Analysis)

Week 7	Classification II (Sentiment Analysis)
Week 8	Reading Week
Week 9	Clustering (Document Retrieval)
Week 10	Clustering (Document Retrieval)
Week 11	Recommendation System
Week 12	Team Project Presentation
Week 13	Team Project Presentation
Week 14	Catch up and Final Review

Assessment

- Class Participation (10%)
- Assignments (50%)
 - Individual assignment (20%)
 - Group Project (30%)
 - Final report
 - Presentation
- Final Examination - 3 hours (CLOSE book)
 - Application of knowledge and skills from lecture materials, and practical knowledge

Class Participation

- Quizzes (5%)
 - Take in-class quizzes via online questionnaire
 - Each quiz may involve 3-5 simple questions related to last lecture
- Finish homework (5%)
 - Submit your homework on iSpace
- No attendance sheet and roll call

Deadlines for all submission are **Saturday** of the week

SEP		1	2	3	4	5	6	
	7	<u>8</u>	9	10	11	12	13	1
	14	15	16	17	18	19	20	2
	21	22	23	24	25	26	27	3
	28	29	30					4
OCT				1	2	3	4	
	5	6	7	8	9	10	11	4
	12	13	14	15	16	17	18	5
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NOV							1	
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	30							
DEC		1	2	3	4	5	6	12
	7	8	9	10	11	12	13	13
	14	15	16	17	18	<u>19</u>	20	14
	21	22	23	24	25	26	27	

← Group Formation Due (ASAP)

← Individual Assignment & Group project Distribution

← Individual Assignment Due

← Group Project Due

Team Project

- **Team Formation**

- Each group can have **up to 6 members**.
- If you have issues with organizing or finding a group, you may consider contacting Assistant Instructor for some help.
- Organizing & finding a group is a part of the required class activities ultimately you are responsible for.

- **General Instruction**

- Define the goal of your project;
- Find your own dataset;
- Form a data analysis team (4-6 members);
- Apply any machine learning algorithm to achieve the goal;
- Present your work and write a final report.

- **More details will be displayed in the Guidelines for Group Project on iSpace**

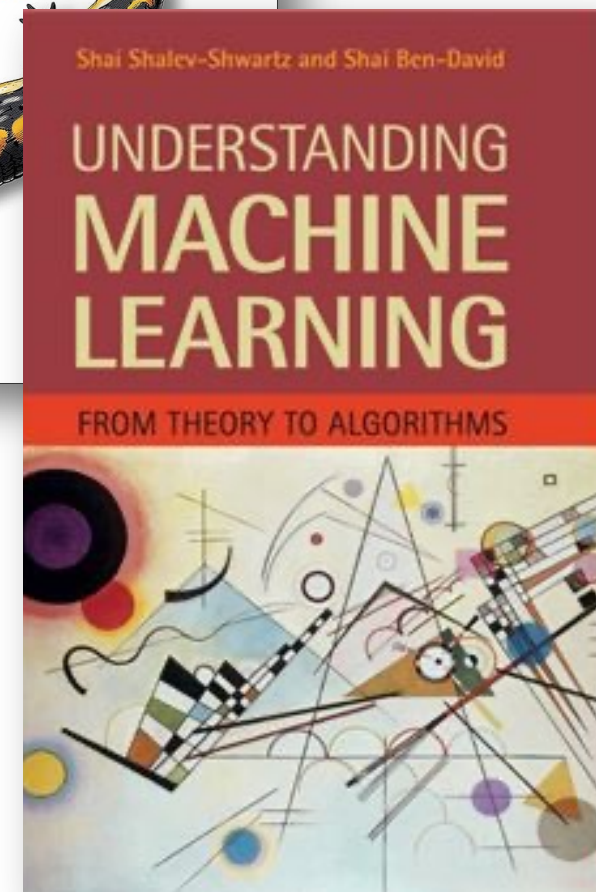
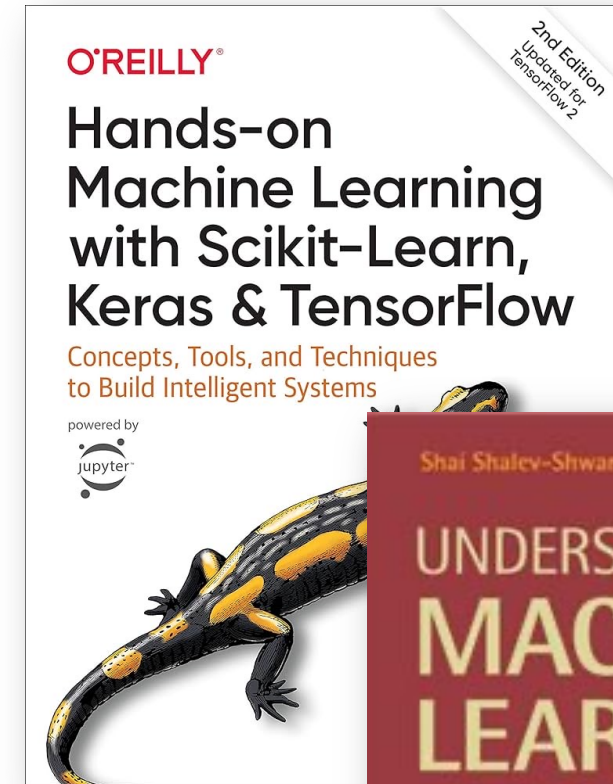
Reference Materials

Textbooks

- [1] Géron, A. (2019). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (2nd ed.). O'Reilly Media.
- [2] Shalev-Shwartz, S., & Ben-David, S. (2014). Understanding Machine Learning: From Theory to Algorithms. Cambridge University Press.

Python Basics

- [1] Introduction to Python (W3Schools):
https://www.w3schools.com/python/python_intro.asp



Question & Answer ?

